

Human - Computer Interaction

NLP-Driven Theater Ticketing Application

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ABSTRACT

The evolution of mobile technology has significantly transformed service accessibility, enabling seamless interactions between users and service providers. This project presents the development of a text-based dialogue system designed to facilitate communication between theater-goers and the box office of a hypothetical theater. The system supports natural language interactions in English, allowing users to obtain show information, book or cancel tickets, and request customer support through a user-friendly interface. To ensure usability and efficiency, the application integrates both manual selection menus and an NLP-powered chatbot, ensuring accessibility for a diverse range of users.

You can view our video here:  Cycle 3

You can view our source code here: <https://github.com/MariaSchoinaki/Thea>

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Introduction

Project Overview

Our theater comprises two auditoriums, each showcasing different productions on various days, with two performances daily, one in the afternoon and another in the evening. To ensure a superior customer experience, the development of an intuitive and efficient system is crucial for managing ticket reservations and addressing inquiries while seamlessly adapting to a rotating schedule of shows.

Spiral Model

This project follows the spiral model of software development, progressing through three iterative cycles:

- **Cycle 1:** Prototyping and user research, where user needs are analyzed, and initial prototypes are created.
- **Cycle 2:** Advanced prototyping and usability testing involve refining functional digital interfaces through iterative improvements driven by user feedback.
- **Cycle 3:** Final implementation and comprehensive evaluation, where the system is deployed on a mobile platform with iterative improvements from extensive user testing.

By adopting an iterative, user-centered approach, this project aims to develop a robust, interactive theater ticketing system, demonstrating the effectiveness of natural language dialogue interfaces in optimizing ticketing and customer support services. The findings contribute to the broader field of intelligent conversational agents and human-computer interaction in service-oriented applications.

1. Cycle 1: Preliminary Assessment

1.1 Target Group

During the initial phase of our interface development following the spiral model, we chose to focus on two specific user groups.

- **1. Individuals who are familiar with technology.**
- **2. Individuals with advanced age.**

The requirements of the first user category are effectively met by strictly adhering to established user interface design principles, which we carefully implemented. The second user group, consisting of elderly individuals, requires a simple and highly accessible interface with large buttons, clear fonts, and minimal complexity to ensure ease of navigation. Additionally, the system must include error-prevention mechanisms and optional human assistance to accommodate users who may struggle with technology.

1.2 Functionality

The initial cycle of prototyping includes the following core functions:

- **1. Displaying information about theater plays**
- **2. Booking tickets**
- **3. Canceling tickets**

These functions were prioritized as they are fundamental to the application's purpose and are expected to yield valuable insights through evaluation methods. More specifically this choice was taken, in order to provide all users a simple interface to avoid confusion and to facilitate the process of buying or canceling a ticket (*either by doing it manually or by using the virtual assistant*). The gathered feedback will guide improvements in the interface for the next development cycle.

1.3 Mockups

Instead of using hand-drawn prototypes, we chose to create basic prototypes with Draw.io, as it serves the same purpose while offering a clearer and more polished visual representation. These prototypes visually represent our initial concept for the user interface, showcasing key screens and interactions. This method enables us to efficiently convey our design ideas and collect early feedback in a clear and structured way.

You can find all our mockups here: [📺 Mockups](#)

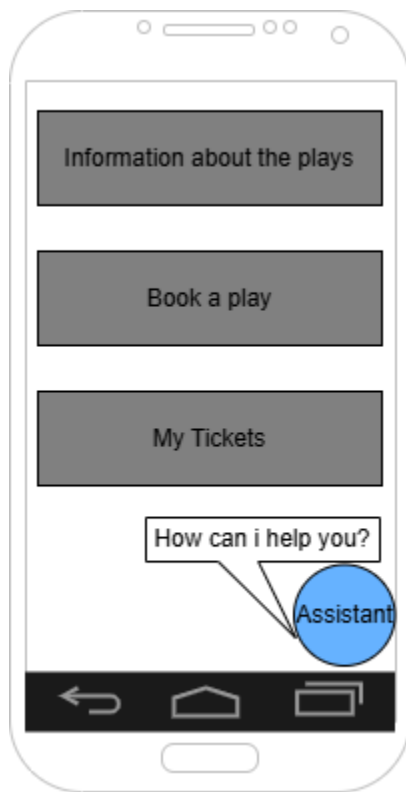


Fig. 1.3.1: Home Screen



Fig. 1.3.2: Booking a ticket with the virtual assistant

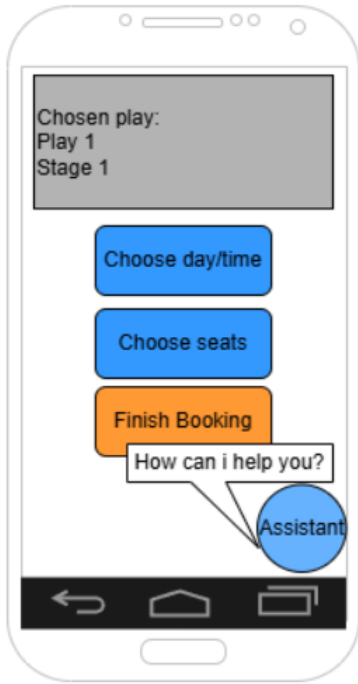


Fig. 1.3.3: Booking a ticket

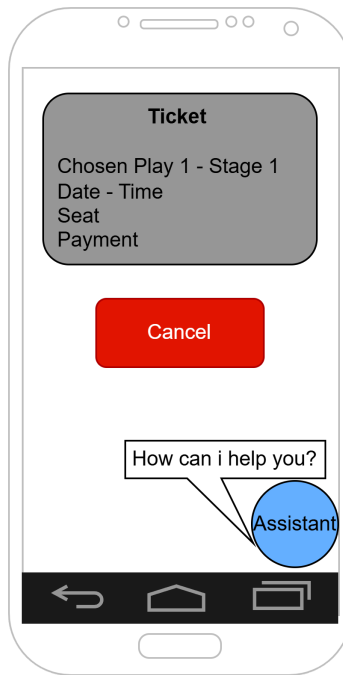


Fig. 1.3.4: Ticket Cancellation

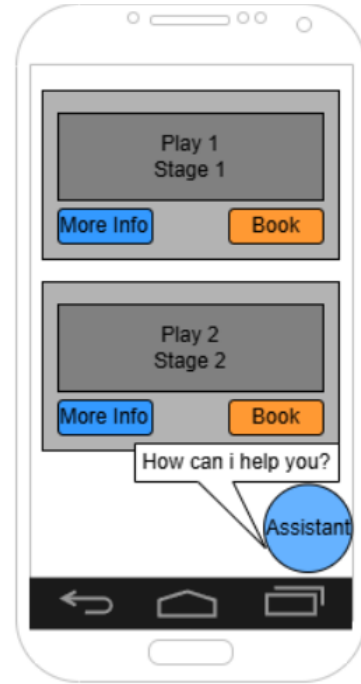


Fig. 1.3.5: Displaying Info about plays

1.4 Evaluation Methods

1.4.1 Questionnaire

A structured questionnaire in Greek was performed, as part of the first cycle of the **spiral development** model to assess the user-friendliness and the perceived effectiveness of early stage prototypes. The aim was to gather both **quantitative** and **qualitative** feedback on the initial user interface, assistant behavior, and expected features of the app (*both closed and open-ended questions*). A total of 18 users completed the surveys (*10 female and 8 male*). The demographic area includes users between the ages 18 and 56+, with different levels of educational and technical backgrounds. It is important that the majority (*100%*) use smartphones **regularly** in their daily lives, but the **level of comfort** differed with advanced features.

As part of the survey, participants were asked if they had encountered any issues when booking tickets or using other applications for customer service (*Fig.1.4.1.1*). This question is intended to provide insight into **user frustration** using existing systems and help us design a more resistant

and user-friendly interface. Future iterations plan to improve assistant behavior by giving clear and descriptive feedback when user inquiries are misunderstood. This reduces confusion and increases **transparency**. After some failed attempts, the assistant will provide the option to **connect directly** to the human representative to ensure that the user's requests are effectively improved. After all, important information such as seating, ticket prices, booking status and more for the entire user trip will be **visible prominently**, with mandatory details always visible and easy to access.

One of the most important findings from the survey was that the **majority** of users (61.1%) **never used** an app to book a theater ticket (*Fig.1.4.1.2*). This outcome presents a valuable opportunity to address the users' need for theater booking. Users are generally active smartphone users, but theater areas clearly **lack familiarity** and exposure to digital solutions. This lack of experience underscores the need for a **well-designed, accessible** and **user-friendly** theater ticket booking application. It allows users to introduce digital tickets in a way that is simple, intuitive and reliable. The purpose of our application is to fill this gap by providing a **guided conversational interface** that feels **natural** and **accessible**, especially for **target groups** that are less technically experienced.

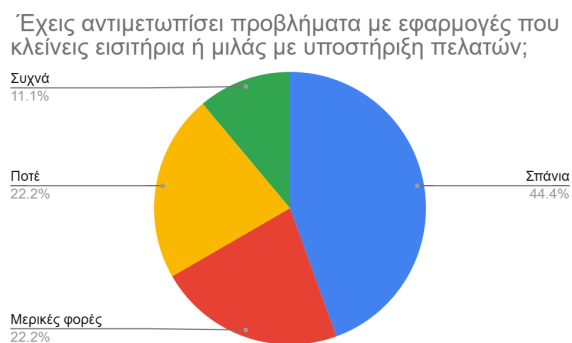


Fig. 1.4.1.1: Answers to the question; “Have you encountered problems with apps when booking tickets or talking to customer support?”

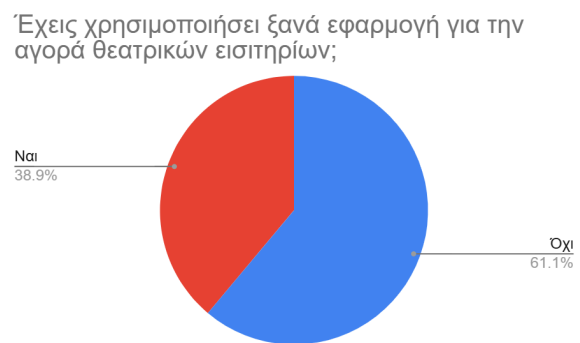


Fig. 1.4.1.2: Answers to the question; “Have you used an app to buy theatre tickets before?”

The results of the questionnaire indicate that the **majority** of the respondents felt **sure** about how to **navigate** through the home screen of the application (Fig.1.4.1.3), with most stating that they **understood the core functions** (such as booking, chatting with a virtual assistant and basic navigation) (Fig.1.4.1.4). Although many users found these two elements very clear, there was a number of users with moderate understanding in the navigation or the core functions, demonstrating that **further refinement** (such as additional guidance), could help improve the experience of the application's users.

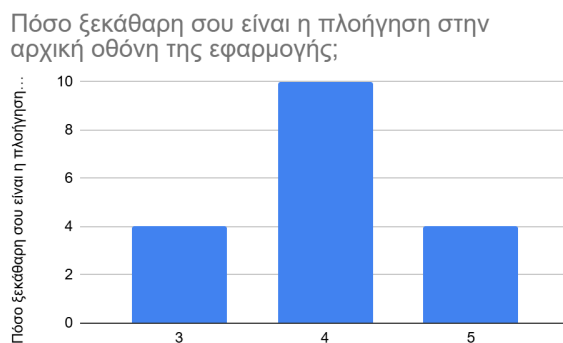


Fig. 1.4.1.3: Answers to the question; “How clear is the navigation on the app's home screen to you?” (1=not clear, 5=very clear)

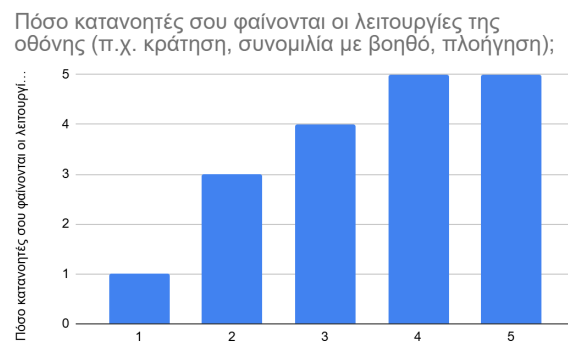


Fig. 1.4.1.4: Answers to the question; “How clear do you find the functions on the screen (e.g. booking, talking to an assistant, navigation)?” (1=not clear, 5=very clear)

Additionally, the feedback on the layout of the information (including buttons, labels, and icons), was generally positive, with most participants expressing **satisfaction** (Fig.1.4.1.5). However two-thirds still answered that they would like our **home screen to have enhanced functionality**, which would translate to adding more information in the layout, in order to support more **functionalities** directly through the main screen (Fig.1.4.1.6).

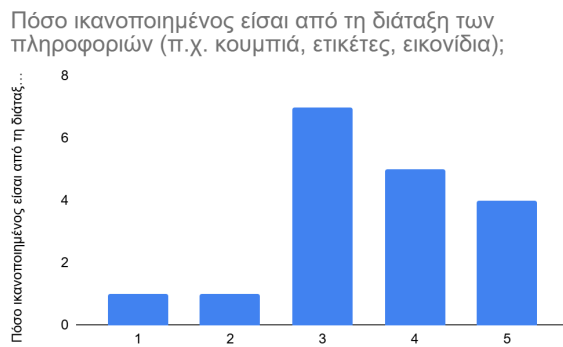


Fig. 1.4.1.5: Answers to the question; “How satisfied are you with the layout of the information (e.g. buttons, labels, icons)?” (1=not satisfied, 5=very satisfied)



Fig. 1.4.1.6: Answers to the question; “Would you like to be able to do more actions from the home screen?”

Another important aspect of the survey focused on user **language preferences** in interaction with applications. Only **16.7%** of those surveyed preferred to use the app only in Greek. In contrast, **50%** of English and an astonishing **33.3%** chose their desire for bilingual support. Based on this data, we have concluded that to ensure accessibility and comfort for a larger audience, the app should support **English** from the first version.

To sum up, while the **majority** of the responders seemed to be **satisfied** using the application, there was a very **small minority** that showed **difficulty** not only in using the **app**, but also in **interpreting** what was happening in different situations in the app shown through images and therefore having a mild first impression about it. The responses gathered, showcase the **importance of improvements** in our application to satisfy both the **tech-savvy** group and those who may **require more explicit guidance**.

You can find our questionnaire results here: [📄 1st_cycle_questionnaire_results](#)

1.4.2 Interviews

To supplement the questionnaire, we conducted interviews with three individuals: a **youth theater producer** and **two retired primary level knowledge individuals**. This was done to

include both digitally literate and less tech-savvy users, in alignment with the target groups defined in 1.1.

Below are some of the questions asked:

- Have you ever used other applications to book online theater tickets? If so, what was your experience with them?
- What was your experience with the Thea prototypes? Was there anything that confused you in the app?
- Did you find the app friendly and easy to understand? Did you need more guidance during your navigation in the app?
- Have you ever used an app with an integrated assistant? Did you understand the role of the assistant? If not, would you like someone to explain it to you?

Based on their responses, we identified **key insights** regarding their experience with the **Thea prototypes**.

Our interviewees generally have limited experience with applications, with some only using the basic pre-downloaded apps of a smartphone and occasionally experimenting with others. While there is some familiarity with purchasing tickets online, they have **not explored theater apps**, suggesting either a **lack of need** or a **preference** for more **traditional methods**.

The app received both **positive** and **neutral** feedback, with some describing it as good and others as average. The ability to **book seats** and **purchase tickets** was considered **useful**, while some users did not find any particular **noteworthy** features. Although some interviewees **did not** encounter **difficulties**, others mentioned that they would prefer a **simpler** and more **user-friendly** experience.

The **information layout** and **visual structure** were generally perceived as **easily understandable**. However, some interviewees **struggled** to **navigate** the app, indicating that while the usability of the app might be enough for experienced users, **additional guidance** could be **helpful** for those **less familiar** with similar applications.

Some interviewees **did not** find the **assistant helpful** and easy to use. Some of those **did not know** what the **assistant** was, and brought up the **need** of having to **explain** its role. As a result, they explicitly mentioned wanting a way to connect directly to a **human employee** in situations where the assistant is **unable** to respond correctly. This feedback aligns closely with the system requirement for **fallback support** and will inform its development in the next cycle.

The **ticket booking** process felt **easy** for **some** but **challenging** for **others**, depending on their level of **experience**. Some interviewees wanted more guidance, which suggests that additional information, such as **step-by-step instructions** or **pop-up messages**, could **improve** the experience.

Another unresolved request was to include **voice input**. Some users, especially those with limited skills, said they would feel more **comfortable** if they could interact with **voice commands** with their assistants. This demonstrated the potential of voice assistants as a more **integrated** mode of **interaction**, especially for users without digital education.

Finally, one additional insight was the clear request to have a place in the app to leave **complaints** and **comments**, especially if something went wrong during the **booking process**. The ability to easily **send feedback** and **message problems** without any issues can not only increase **transparency**, but also increase **trust** in the system.

1.4.3 Summary

From the questionnaire and the interviews, we summed the following observations/insights:

Insights:

- **Low digital familiarity:** Participants, especially retirees, had only experience with digital apps that did not go beyond basic pre-installed features. This indicates the need for low complexity interfaces.
- **Desire for simplicity:** The elderly emphasized the need for simpler, beginner-friendly designs, especially for core features such as navigation and booking (*larger texts and clearer navigation*).

- **Minimal exposure to theater-specific apps:** None of them used an app specifically for theater tickets.
- **Various feedback on usability:** Some have described the app as “good” and found important features useful, while others found a lack of average or distinctive features.
- **Demand for human interaction:** Several respondents said they prefer to be forwarded to an actual employee if their assistants are not providing the right answer. This highlights the need to implement fallback options for human support.
- **Assistant confusion:** The elderly were uncertain about the assistant’s purpose or how to use it, highlighting the need to explain its role or offer onboarding prompts (*fAQ*).
- **Voice-based interaction:** For users who face challenges typing or navigating complex menus, a voice assistant was proposed as a more inclusive alternative for completing common actions, such as booking or requesting information.
- **User complaint submission:** A few users expressed the need for a way to submit complaints or feedback within the app, which would make the system feel more complete and responsive to user concerns.
- **Help & guidance expectations:** Participants indicated that having more in-app support (*e.g. pop-ups, tutorials, or a help button*) would make the experience more comfortable and less intimidating.

The results of the interviews **increased** many of the observations from the survey, but certain **user-friendly** concerns were added to the **depth** and **context**. Most importantly, it shaped **key decisions** about **onboarding**, **assistant integration**, and **error-handling mechanisms** so that the app is **comprehensive** and **accessible** to a wide range of users **regardless** of **technical literacy**.

Based on these findings, the **second cycle** of the spiral model will focus on implementing and evaluating the additional functionalities such as voice interaction, improved assistant transparency, complaint submission, direct human support access etc all of which emerged

directly from this user feedback and the additional required support functions such as paying during ticket booking.

2. Cycle 2: Prototyping

2.1 Target Group

In the second development cycle, we maintained our focus on the two primary user groups defined in **Cycle 1**:

1. **Users familiar with technology**
2. **Elderly users with limited digital experience**

Although **visually impaired individuals** were not explicitly targeted as a separate group, we observed that many elderly users face age-related visual challenges. Therefore, designing for accessibility and ease of use for older adults resulted in addressing some of the needs of users with reduced vision as well.

For example, adjustments such as larger fonts, voice interaction features, and simplified navigation were introduced to serve the elderly audience, but these improvements also happen to benefit partially sighted users. This alignment reinforces the inclusive value of our design decisions, even if it wasn't the direct objective of our second cycle.

2.2 Functionality

The prototypes developed in **Cycle 2** retain all functionalities from **Cycle 1**, while introducing several new features as well:

- **4. Communication with a human employee**
- **5. Payment integration (*ticket booking process*)**
- **6. Manage booked tickets (*view, sent, cancel, edit*)**
- **7. Submit complaints or feedback (*report issues or express concerns*)**

- **8. Addition of two DCG grammars.**

To demonstrate the chatbot's natural language understanding, we created two Define Clause Grammars (*DCG*) in Prolog.

A. Simple Request Grammar.

This grammar accepts only sentences in the form of Subject + Request + Action. We have predefined a set of requests (*"want to"*, *"wish to"*, *"would like to"*) and actions (*"book a ticket"*, *"cancel a reservation"*, *"get information about the theater"*, *etc.*), which makes this grammar ideal for a first prototyping.

B. Semantic Grammar.

This grammar breaks each sentence into noun phrases, verb phrases, modifiers and adverbs. It covers a big range of combinations of sentences, by using techniques like synonyms (*e.g. "book" and "reserve" and "buy"*) and nested modifiers (*e.g. "for Hamlet by Shakespeare"*). With this grammar we tried to implement the logic of the chatbot to understand the intent of the user.

```
?-  
% c:/Users/giwrg/Downloads/testprolog.pl compiled 0.00 sec, 4 clauses  
?- phrase(s, [i, want, to, book, a, ticket]).  
true .  
  
?- phrase(s, [i, would, like, to, cancel, a, reservation]).  
true .  
  
?- phrase(s, [i, wish, to, leave, a, complaint]).  
true .  
  
?- phrase(s, [i, want, to, get, information, about, the, theater]).  
true .  
  
?- phrase(s, [i, want, to, book, a, flight]).  
false.  
  
?-
```

Fig. 2.2.2: Screenshot of a simple Prolog DCG grammar with hardcoded requests.


```

?- phrase(s(Sem), [i, want, to, cancel, a, reservation]).
% c:/Users/giwrg/Downloads/thea_grammar.pl compiled 0.00 sec, 54 clauses
?- phrase(s(Sem), [i, want, to, cancel, a, reservation]).
Sem = request(speaker, cancel, object(reservation)) .

?- phrase(s(Sem), [i, would, like, to, book, a, ticket, for, hamlet, by, shakespeare, for, tomorrow]).
Sem = request(speaker, book, [object(ticket), play(hamlet), author(shakespeare), date(tomorrow)]) .

?- phrase(s(Sem), [i, want, to, get, information]).
Sem = request(speaker, get_info, object(information)) .

?- ■

```

Fig. 2.2.2: Screenshot of a Prolog DCG grammar for semantic representation of the tasks of booking, canceling, and info providing.

2.3 Mockups

Following the initial idea and low-fidelity prototyping in the 1st Cycle, the **2nd Cycle** of development focused on refining our concepts and designing **high-fidelity** mockups that simulate the **key functionalities** of our app. The goal was to bring our envisioned user experience closer to a **realistic** scenario while still maintaining the **flexibility** for **iterative feedback** and **improvement**.

To this end the prototypes were developed using **Figma**, allowing smooth transitions and functionality simulations.

You can find all our mockups here: [Mockups](#)

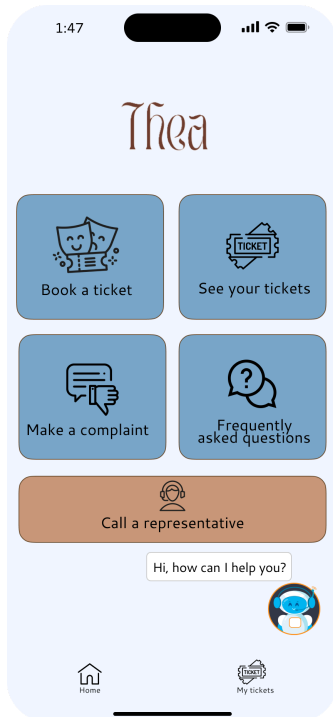


Fig. 2.3.1: Home page with the apps functions

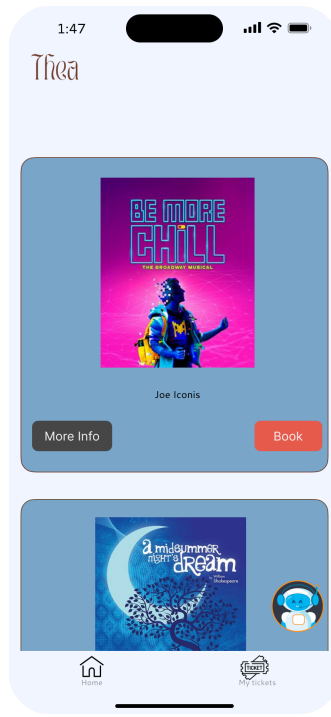


Fig. 2.3.2: Available theater plays

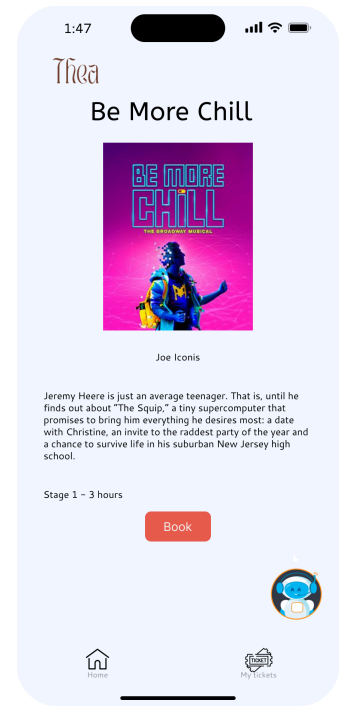


Fig. 2.3.3: More Info about the “Be more chill” theater play

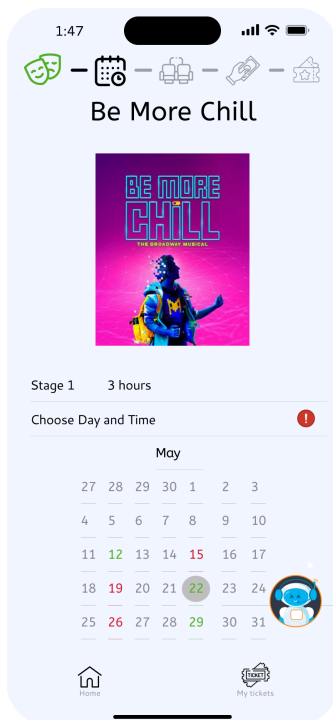


Fig. 2.3.4: Booking page for a play



Fig. 2.3.5: Continuity of booking page (stage booking seats)

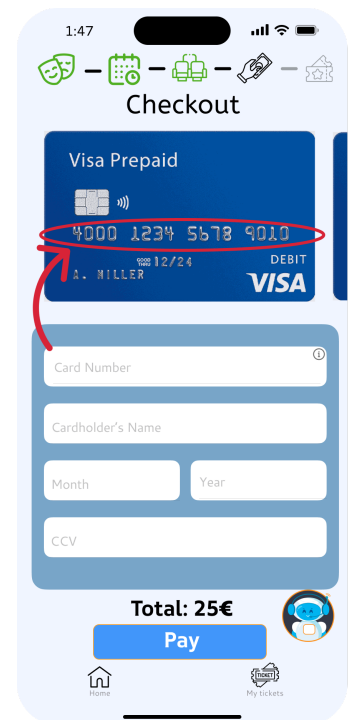


Fig. 2.3.6: Payment page showing information (i) for the Card Number

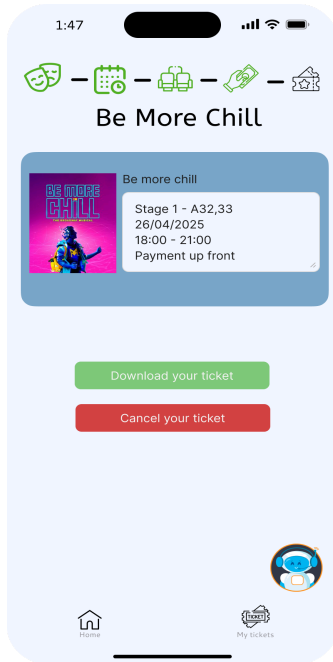


Fig. 2.3.7: Final Stage of booking, showing the ticket

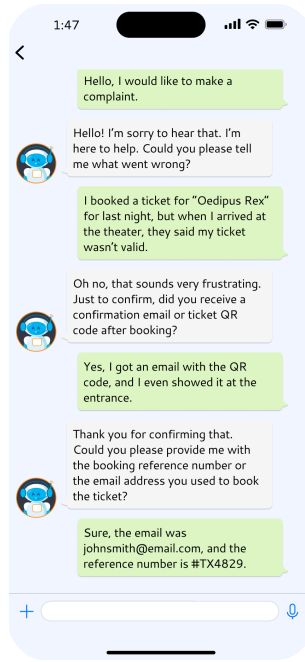


Fig. 2.3.8: User makes a complaint using chatbots assistance

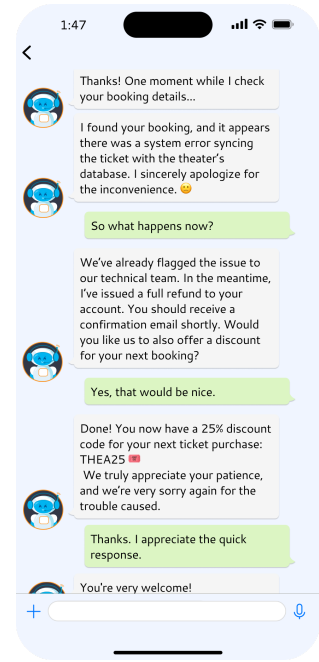


Fig. 2.3.9: Continuity of user's complaint

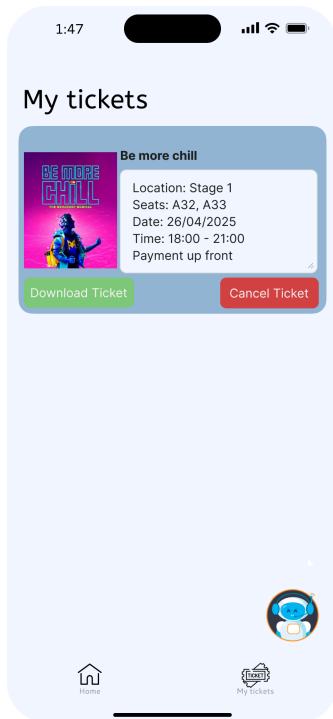


Fig. 2.3.10: My tickets page, showing the purchased tickets

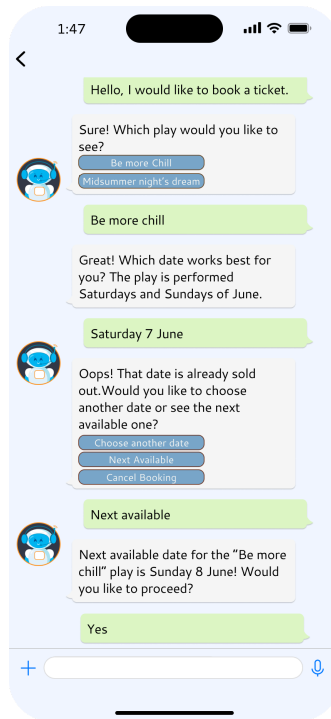


Fig. 2.3.11: Booking ticket process through chatbot with recovery mechanism when errors occur.

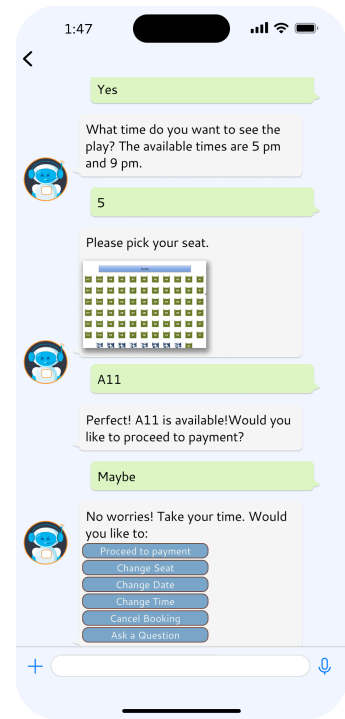


Fig. 2.3.12: Booking ticket process through chatbot with recovery mechanism when errors occur.

2.4 Evaluation Methods

2.4.1 Usability Inspection

During usability inspection interviews, **Thea app** impressions were generally positive. Participants each described their experience as "*easy to use*", "*pleasant*", and "*clear*". Special focus was noted with regard to design, with the interface elements termed as carefully considered and visually appealing. Overall, the app was considered **functional** and **fairly well-designed** for its intended purpose.

But also some practical advice and criticism resulted from the discussion, and some of the points, where the **user experience** can be **improved**, were highlighted. One of the most mentioned points of criticism was concerning the **screen of the integrated chat assistant**. Adding a **title** at the top of the chat and a default **user picture** could contribute to the feeling of an **active** and **individualized** conversation.

For the presentation of theater plays, usability assessors commented that the text size in the “**More Info**” screen is too tiny and hard to read. Further additions suggested are as below:

- Adding **labels** to the details of each play (*e.g. Duration: 3 hours, Stage: Stage 1*)
- The **availability period** for each play
- Clear display of the **seats** selected by the user

The cancellation of the booked ticket was another essential part. When selecting “**cancel your ticket**”, it was suggested that the user is automatically **redirected** to the **refund page** instead of having to navigate to it. At the same time, the addition of **confirmation notifications** for **important actions** was considered indispensable, to inform users that their operations were successfully *completed* or *not*. Additionally, the inclusion of a **reschedule option** for **reservations** was considered as a way to make things more **flexible**.

One particularly practical suggestion was put forward regarding the “**My Tickets**” section. It was proposed by inspectors that all tickets be made interactive so that, with a tap, a **QR code** or **barcode** would appear, allowing for **quick** and **easy access** into the venue.

Last but not least, another point of usability was noted: the app would be improved by adding a **horizontal scroll arrow**, particularly in areas with content **extending beyond** the visible screen. This would enhance **navigation**, especially on smaller devices.

To sum up, according to the **usability reviewers**, the **Thea app** already demonstrates strong **design principles** and **solid usability**. With **targeted improvements**, such as enhancing the **chatbot experience**, making **key information easier to find**, and **incorporating practical features** like **QR code** access to **tickets** or **streamlined refund options**, the **overall user experience** could be **significantly improved**. These enhancements would further **increase** the app's **usefulness**, **clarity**, and **accessibility** for all **target user groups**.

2.4.2 Talking Subjects

To further assess our app's usability level, we used the "**talking subjects**" evaluation method. To complete this, we asked a **computer science student**, who was provided with the mockups of our app, to complete three simple tasks: **booking a ticket**, **submitting a complaint**, and **canceled a ticket**. While carrying out the tasks, the participant was requested to **verbalize his thoughts** and provide **real-time feedback** on his progress and the problems he may have encountered. Below is an **overview of the tasks** completed (*where the participant will be referred to as "user"*) and the **main findings** that came from them.

Task 1: Ticket Booking Process

The user began at the "**Home**" screen and easily spotted the option of booking a ticket. However, he stated that he would navigate to the "**Available plays**" page because till now, he has no info about the plays offered by the theatre. Being in the available plays section, the user browsed the plays and their descriptions by using the "**More Info**" option. Once he found a play appealing to him, he then proceeded by selecting the "**Book**" option. During the booking task, the user progressed through the following steps:

- Selected the desired **date and time**
- Scrolled and chose his seat from the **seating plan**

- Clicked the **“Book” button** to continue

After clicking the **“Book” button**, he expected to **be taken to the payment section**, where he **filled in his details** and **confirmed the payment**. Having done this, the ticket was **available to download**. He also noted that he would navigate to the **“Home” screen** and choose **“See your Tickets”** for a final **confirmation** of the ticket booking.

Main Findings:

The user **expressed confusion** during the booking task because he thought that the stage and duration information (**Fig .2.4.2.1**) before choosing the day and time were meant to be selected. Also worrying was the fact that the user didn’t choose the **“My Tickets” option** in the navigation menu in order to confirm the booking of the ticket but preferred to first navigate to the main screen and then to the **“See your Tickets” section**.

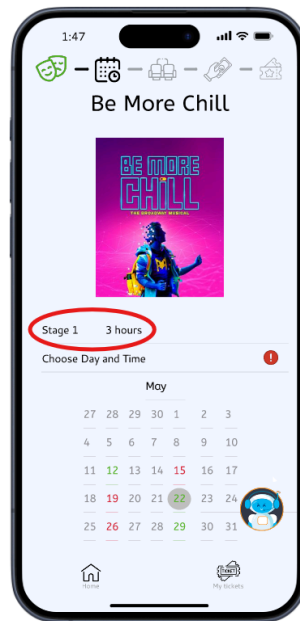


Fig. 2.4.2.1: The mockup showcasing the point where the user got confused into thinking that he must select “Stage 1” and “3 hours”.

Task 2: Submitting a Complaint

Once again, the user started from the “**Home**” screen. Due to the fact that he **didn’t see any option** that could help him submit a complaint, he wondered if he could **use the assistant** to file it. Therefore, he opened the chat bot, and started typing the complaint.

Main Findings:

Although the user was able to use the assistant to perform this task, he **did not fully understand** that complaints were exclusively handled through the chatbot. This **uncertainty** indicates a **need for further clarification** about the functionality that the assistant supports.

Task 3: Ticket Cancellation Process

This time the user stated that he would use the “**My Tickets**” option in the navigation menu in order to **find a cancellation option**. He was anticipating that in case the cancellation option did not exist in the “**My Tickets**” section, he then would **proceed using the assistant**. Additionally, he stated that he would prefer to perform this task “**manually**” and if he needed any further assistance he would use the chat-bot.

Main Findings:

This task outlined only **one problem** that exists and it was also mentioned by the user: the **lack of providing** the users with the ability to **modify their tickets**. Apart from this, the user did not face any other problems, with the cancellation process being **very clear on how to complete**.

To sum up, this evaluation method determined that while the app’s navigation **was in general intuitive**, in some cases **interface clarity improvements** (*such as labeling, adding more manual functionality and better showing the assistant’s role*) were necessary in order to **avoid confusion**.

2.4.3 Cognitive Walkthrough

To better understand how a **first-time user** would interact with our app, we ran a “**cognitive walkthrough**” based on the user journeys we had in mind. The goal was to **evaluate** if the **interface** naturally **guides** the user through key tasks, especially **without previous experience**.

We focused on three main scenarios and asked ourselves:

- Will the user **understand** what to do?
- Will the options on the screen feel **obvious**?
- Is it **clear** what each action does?
- Will users get some sort of **feedback** that they did the right thing?

Scenario I: Booking a Ticket

“The user wants to book a ticket for a theater performance.”

Actions:

1. Home Screen

The user starts his **navigation** through the “**Home**” screen where he sees the clearly labeled **function buttons**.

- They can easily choose the button of “**Book a ticket**” to start the process.
- The option **clearly** communicates what **process** it starts.

2. Available Plays Screen

The user lands on the “**Available plays**” screen and spots the play they’re **interested** in. Right below each play, there’s a clearly labeled “**Book**” button.

- They **easily** spot the play they’re interested in and click “**Book**”.
- The option is **visible** and **clearly** communicates what **action** it performs.

3. Details Screen

The user gets a short **description** of the play, including the **stage** and its **duration** in the “**Details**” screen.

- They press “**Book**” again to move **forward**.
- This **reinforces** their **decision** and keeps the **flow** moving **smoothly**.

4. Date & Time Selection

Now, the user lands on the **“Book” screen** and picks a **date** from a horizontal **calendar** and chooses between **afternoon or evening** performances.

→ The design is **simple**, and the **visual separation** of steps (*date/time, seat etc.*) helps **avoid confusion**.

→ The screen **clearly** shows what comes **next**.

5. Seat Selection

In the same **screen**, a **visual seat map** appears while scrolling. **Green** seats are **available**, **red** are **taken**.

→ The user taps the **seats** they want and hits **“Book”**.

→ The **color cues** and **layout** make it almost **instinctive**.

6. Checkout Screen

The user now sees the **“Checkout” screen** with card fields and the ticket price.

→ They **fill out** payment info and click **“Pay”**.

→ **Standard** credit card **layout**. No surprises.

7. Confirmation

After paying, the user is taken to the **“Confirmation” screen**.

→ They see a **summary** of the **booking** and can **download** or **cancel** their **ticket**.

→ Clear **feedback** on what just happened, with **further control** if needed.

Scenario II: Cancel a Ticket

“The user decides to cancel a ticket they previously booked.”

1. Home Screen

The user starts on the **“Home” screen**, where **clearly** labeled **navigation** buttons are visible.

→ They select **“See your tickets”**, which **clearly** communicates the **function**.

→ The label is **prominent** and **easy** to understand.

2. My Tickets Screen

The user lands on the **“My Tickets”** screen, which shows **active** bookings with the performance title, date, and time.

→ Each ticket includes a clearly visible **“Cancel Ticket”** button in **red**.

→ The **red color** and **button placement** draw **attention** and signify a **critical action**.

3. Cancellation Confirmation

After tapping **“Cancel Ticket”**, a **confirmation prompt** appears:

“Are you sure you want to cancel this ticket?”

→ This prevents **accidental cancellations**.

→ The user **confirms** by tapping **“Yes, cancel”** or can **back** out with **“Go back”**.

4. Feedback & Visual Update

Once **confirmed**, the app provides instant **feedback**:

→ A **message** such as **“Your ticket has been canceled”** appears.

→ The ticket **disappears** from the list or is marked **clearly** as **“Canceled”**.

→ This gives the user **clarity** and **peace of mind** that the **action** was **successful**.

Scenario III: Contacting a Representative

“The user wants help from a real person.”

1. Home Screen

The user begins at the **“Home”** screen, where core options are laid out.

→ They tap the **“Call a representative”** button.

- The button is styled in a **contrasting color** to stand out from other actions.
- Clear labeling **avoids confusion** with chatbot support.

2. Contact Options Screen

Tapping the button brings the user to a dedicated **popup contact message**.

- Depending on the system, it either **initiates a phone call** or presents clear contact options (*phone number, email, or request callback*).
- The screen also displays **working hours** and support availability.

3. Confirmation or Feedback

Once the action is taken, the user receives confirmation:

- If a call is initiated, they see “**Calling...**”
- If a message is sent, the app confirms: “**Your request has been submitted. We’ll be in touch shortly.**”
- These **visual cues** provide **reassurance** and **clarity** that the user is **receiving help**.

3. Cycle 3: Final Implementation & Evaluation

3.1 Target Group

The **target group** for the **third** phase of our **user interface development** remains the **same** as in the **first and second phase**. This **consistency** helps us **maintain a continuous approach** to design **enhancements**, keeps **accessibility features a priority**, and ensures we meet the **unique needs** of these **users**. As a result, we can **improve** the overall **usability** and **satisfaction** for our **users**.

3.2 Functionality

No additional features were introduced during this development cycle. Instead, the **existing functions** were **coded** and **refined** to **better suit** the **requirements** of our **target group**.

3.3 Interface

During the **final stage** of developing our user interface, we used **feedback** from earlier reviews to further **enhance** our design. We **transformed** the **detailed mockups** made in **Figma** into **working code**, making sure the **interface** was both **attractive** and **easy** to use in **real-life situations**. Our **main priorities** were to **improve accessibility** and **resolve usability** problems noticed in **previous cycles**, such as making **buttons easier to see** and making **navigation prompts clearer**. Our goal was to create a **smooth** and **enjoyable user experience**.

We've taken all the **feedback** from the **second development cycle** and used it to **guide** the **creation** of our user **interfaces**. Every **aspect**, from the **layout** to the **features**, is **shaped** by what **users** have told us they **want**. Our aim is to design **intuitive interfaces** that make our **products easy** and **enjoyable to use**, with every **choice** based **directly** on **user feedback**.

Below, you can see the **user interfaces (screens)** of our **mobile app**, which are based on the **designs** we previously shared in **Figma**.

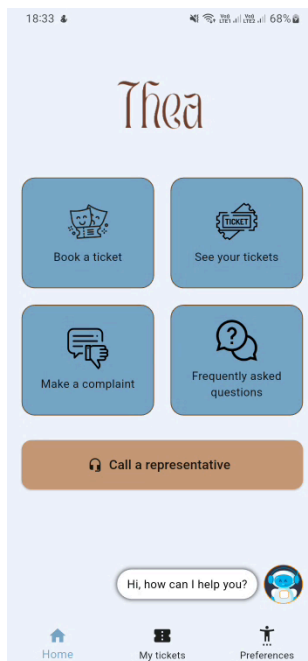


Fig. 3.3.1: Home page with the apps functions

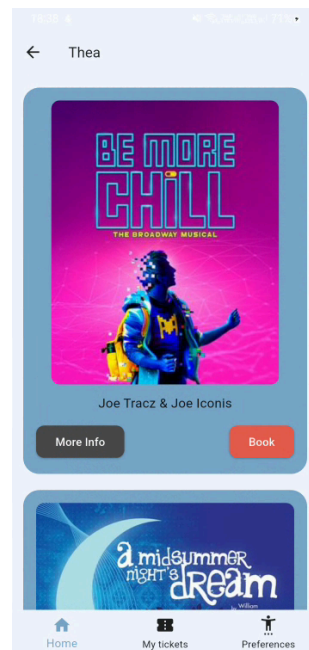


Fig. 3.3.2: Available theater plays

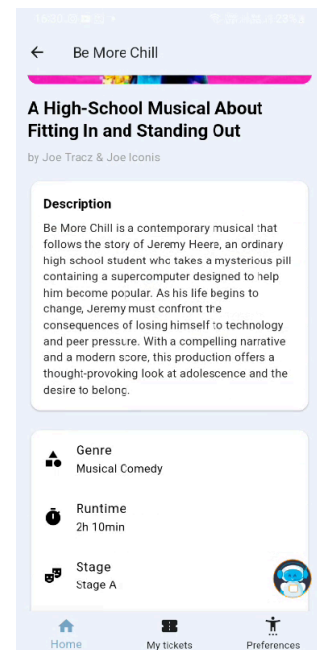


Fig. 3.3.3: More Info about the “Be more chill” theater play

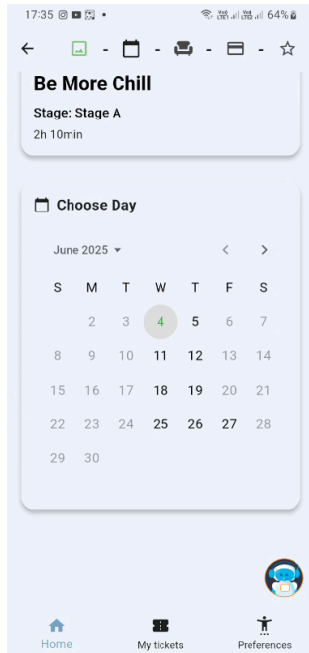


Fig. 3.3.4: Booking page for a play

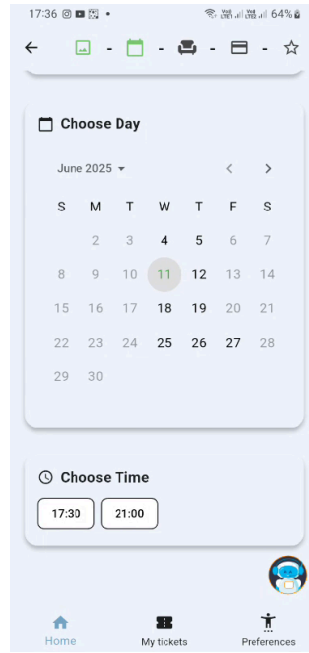


Fig. 3.3.5: Continuity of booking page
(stage booking time)

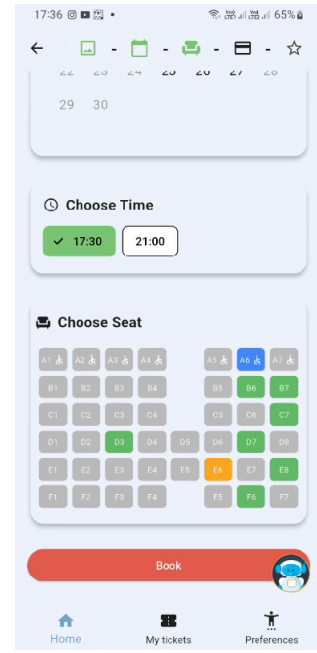


Fig. 3.3.6: Continuity of booking page
(stage booking seats)

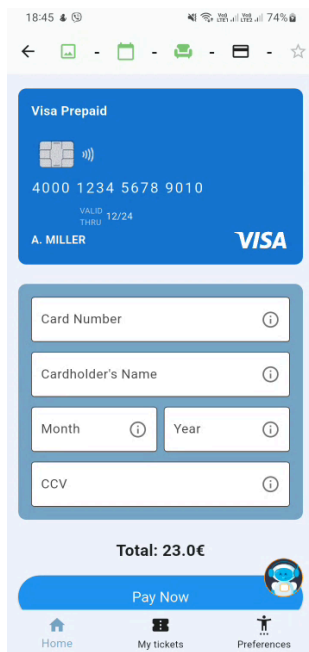


Fig. 3.3.7: Payment page

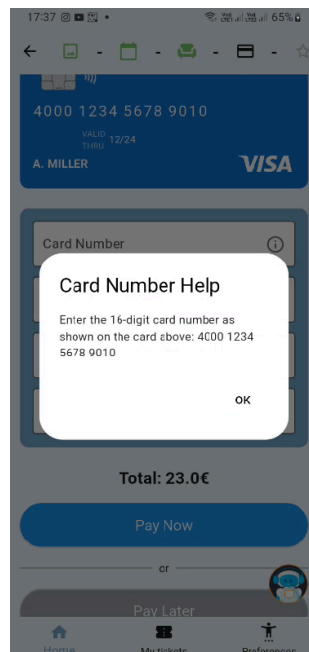


Fig. 3.3.8: Payment page showing
information (i) for the Card Number

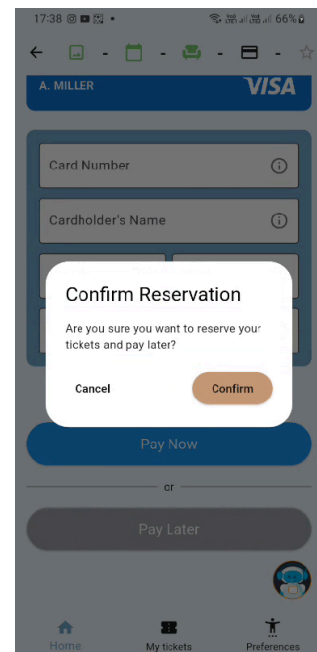


Fig. 3.3.9: Payment page showing
confirmation message

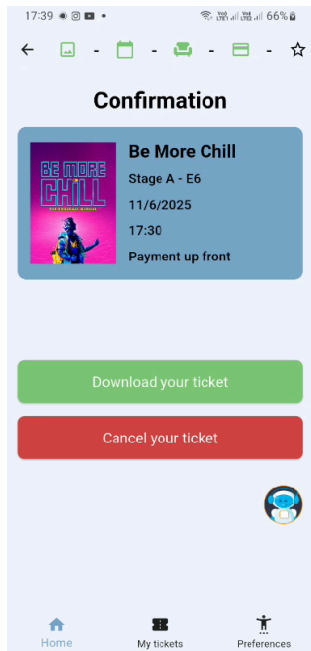


Fig. 3.3.10: Final Stage of booking, showing the ticket

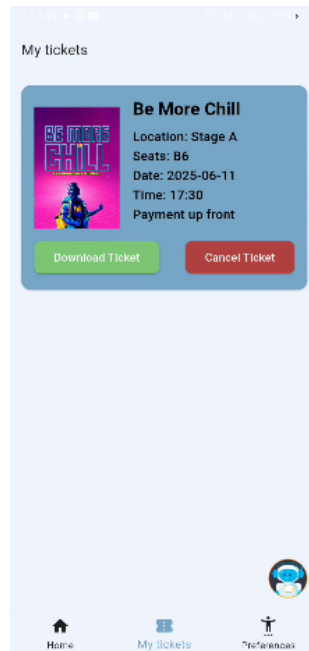


Fig. 3.3.11: My tickets page, showing the purchased tickets

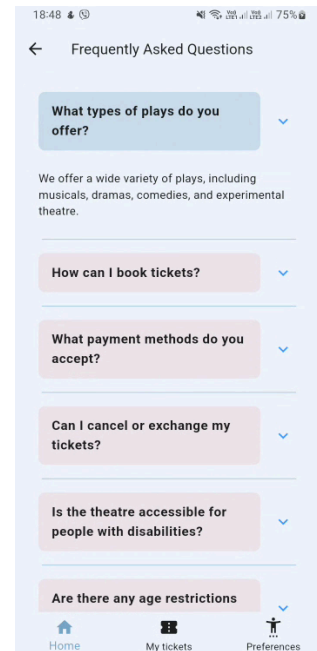


Fig. 3.3.12: FAQ screen

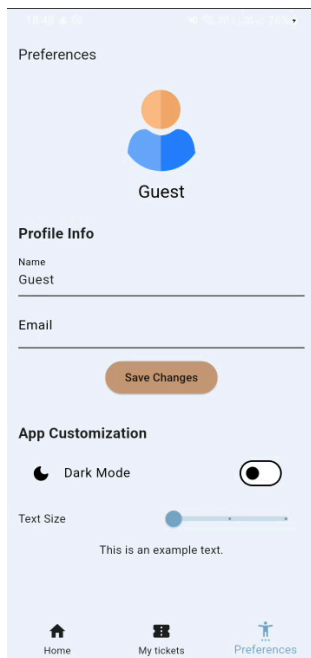


Fig. 3.3.13: User's preferences screen

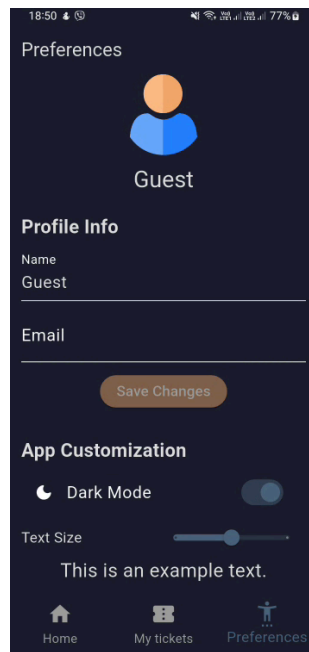


Fig. 3.3.14: User's preferences when selecting dark mode and text size

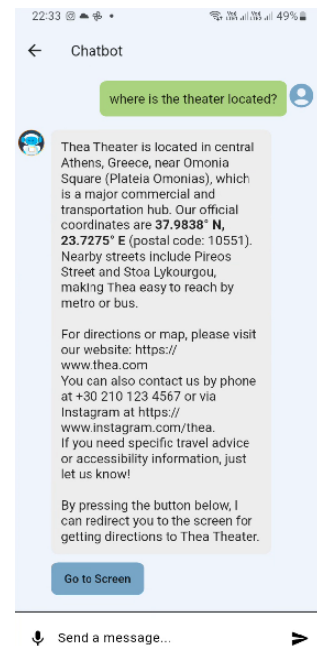


Fig. 3.3.15: Chatbot answering a simple question about the location of the theater

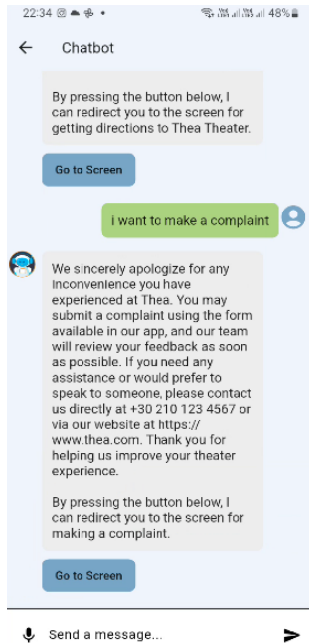


Fig. 3.3.15: Chatbot redirecting user to make a complaint

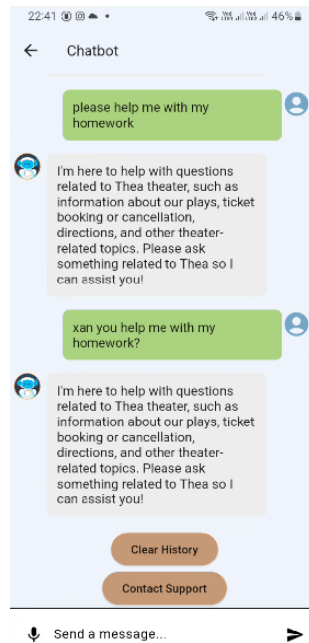


Fig. 3.3.15: Chatbot 3 shot limit tolerance

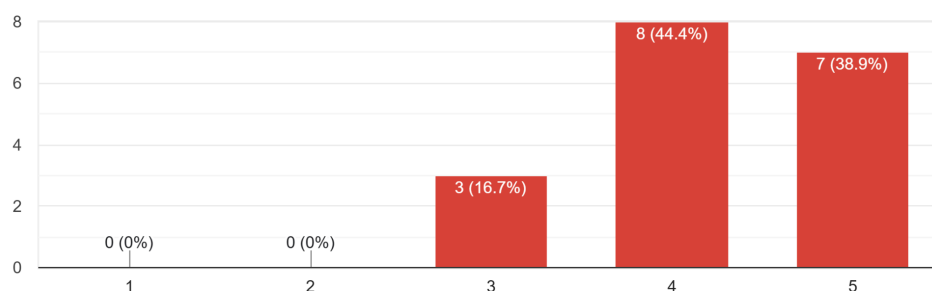
3.4 Evaluation Methods

3.4.1 Questionnaire

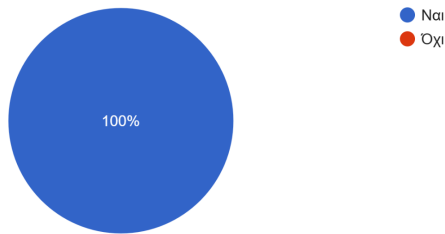
We created another questionnaire in Greek, similar to the first one in the 1st cycle, focused on the app's usability and design. It was completed by 18 respondents, of whom 16 were regular theater-goers and showed interest in a mobile application that would allow them to automate the ticket booking process.

Most respondents agreed that interaction with the app's various screens was straightforward and intuitive, and the chatbot was cited as particularly beneficial and easy to navigate. All participants expressed that they would recommend the app to others.

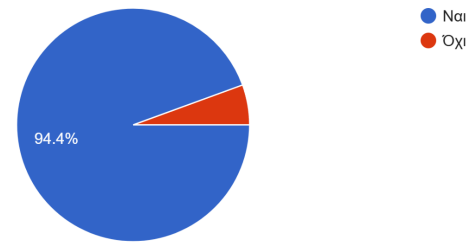
Πόσο κατανοητές σου φαίνονται οι λειτουργίες της οθόνης (π.χ. κράτηση, συνομιλία με βοηθό, πλοήγηση);
18 responses



Είναι σαφείς οι διαθέσιμες επιλογές της αρχικής οθόνης;
18 responses



Θα πρότεινες την εφαρμογή σε άλλα άτομα;
18 responses



You can find our questionnaire results here: [📄 3rd_cycle_questionnaire_results](#)

3.4.2 Usability Inspection

As another evaluation method for our finalized app ,we conducted a **usability inspection**, where participants valued its **polished interface**, **intuitive flows** and **comprehensive features**.

Tapping the chat icon immediately revealed a **clear chat room** interface with a friendly avatar and typing indicators to keep users **informed while waiting** for a response.

One **key info** we got from the inspectors was that there is **no need** for the chatbot and our **manually process app** to do the same thing. Their suggestion was that we should **focus** on one design pattern which after **A/B testing** concluded that it is: **The user can manually do the actions and can use the chatbot as an assistant of how to do them.**

Additionally the theater play details in the “**More Info**” screen were presented in **legible texts**, each clearly **stating its meaning** - *the title was larger and bold, followed by the authors and description and in the end presenting the runtime duration and the stage of the play.*

The participants also appreciated the **ease** of **booking** a ticket. They mentioned that the existence of a **progress bar** showing the current step and **revealing** a step **only after the previous** one is **completed** (like choosing time only after choosing the date), helped **prevent miss-taps** and retain a **serial flow** in the process. Moreover the **marking** of **selected** inputs as **green** and the **not available** as **grey**, the existence of an **organized seat map**, and the display of the **total price**, were considered **positive** and **useful**. Also, in the payment section the display of an **example card** and in each input form the addition of a **help button**, as well as the providing of an option of **paying at the counter desk**, in combination with a **confirmation pop up** about the completion of the process, all added up to the **positive** experience of booking a ticket.

The “**My Tickets**” screen was also positively reviewed, because it provides a **convenient** way to have a **digital wallet containing the tickets**, and supplying the users with the capability to either **download** the ticket to their device (a **pdf file** of the ticket followed by a **QR code**) or **cancel** it.

Lastly they highlighted the **usefulness** of the “**Preferences**” screen, where users **can adjust** the app based on their **likings**.

Overall, the participants found it **effortless** to discover plays, book tickets, manage tickets or chat with our virtual assistant, leaving the inspection **confident** that Thea is **ready for release**.

3.4.3 Talking Subjects

During the Cycle 3 final evaluation, we conducted once again a “**talk-aloud**” **session**, where we asked the participant to **verbalise his thoughts** while completing various tasks. Although the overall **navigation remained “intuitive”**, there were few findings that surfaced and needed to be mentioned.

Booking a ticket.

The participant started from the Home Screen. Then he pressed the book a ticket button to see the available plays and opened a few More Info screens to find something he likes. When he found something that fulfilled his desires, he then proceeded in pressing the book button, in order to be navigated to the booking procedure.

One interesting finding here, is that the participant, because of his **familiarization** with **technology**, and more specifically with **booking ticket systems**, **did not notice** that there was a **progress bar** showing the steps he was completing at that time and the steps that remain and need to be completed. This is interesting because for users with **familiarity** in technology the completion of tasks such as **booking a ticket** is completed **glibly**, but for elderly people (*which are included in our target group*) a **bar** showing the **current** and the **remaining steps** helps **clear out confusion** and **frustration**.

Another point that made the participant pause the completion of the task and ask a question was the option to **pay later**. He questioned why the payment was **not compulsory** on that step. The answer to this is, because even if we live in a digital world where most people possess a **credit card**, there are also people (*and especially elderly*) that **do not own** one and need to also **be included** and make their booking experience as **easier** as possible.

To sum up, everything flowed normally and the participant expressed **satisfaction** while completing the task.

Viewing booked and cancelling tickets.

Starting once again from the home screen, the user stated that he **recognises** two ways to **view** his **tickets**: one being from the big see your **tickets button**, and the other one from the **My Tickets** button in the **navigation bar** at the bottom. This time he chose to navigate through the navigation bar and proceeded to cancel the ticket he wanted.

This task showcased the **clarity** that characterises the process of both **downloading** and **cancelling** the ticket and was completed **smoothly**.

Other actions.

Every other action that was asked from the participant to perform (*e.g. making a complaint, talking to a theatre employee, viewing more info about a theatre play, adjusting preferences like text size*), was completed **smoothly** and **without hesitation**. Throughout these flows the participant **navigated** the interface **confidently**, **recognised** required controls **immediately**, and **did not articulate** any **confusion** or **frustration**.

A **very important observation** that needs to be talked about is the fact that the participant **did not use the virtual assistant** for any of the tasks he was asked to complete. Even though he **acknowledged** its existence, he once stated that he would use it **in case something went wrong** or was **stuck** during a step.

This once again highlights the importance of the **chat bot**. For **non-tech-savvy** users (*like many elderly*) such an option is **necessary** to exist so they can complete tasks **without** having to perform **many actions** - *they just tell the virtual assistant what they want and it does all the work for them*. But even for people that are more **familiar with technology**, the existence of a tool to **help** them in case that something goes **wrong**, is very **prominent**.

3.4.4 Cognitive Walkthrough

To ensure that the **final implementation** of our theater ticketing application is **intuitive** and **user-friendly**, we performed a **cognitive walkthrough** based on **typical user scenarios**. Our **aim** was to determine whether the revised interface **naturally guides** both **first-time** and returning **users** through all **essential tasks**, particularly those **identified** as most **critical** in **previous cycles**.

Key questions guiding our analysis:

- Does the **user** know what to do at **each step**?
- Are options on the screen **obvious** and **understandable**?
- Is the **effect** of each **action** **clear**?
- Does the **app** provide **immediate** and **clear feedback** for **user actions**?

We focused on the **three** primary user **scenarios** that define the **core functionality** of the application.

Scenario I: Submitting a Complaint through the Assistant

"The user wants to submit a complaint or provide feedback about their experience through the chatbot."

1. Home Screen

The user begins at the **"Home"** screen, where the **chatbot bubble** is clearly visible and shows a **robot** icon.

→ The **bubble** is **accessible** and **stands out, inviting** users to interact for **assistance**.

2. Chatbot Screen

The user taps the **chatbot bubble**, which opens a **chat screen**.

→ A **friendly** welcome **message** appears:

"Hello! How can I help you today? You can type your question, submit a complaint, or ask for support."

→ The user **communicates** with the **chatbot**, either by **typing** or **using speech**, to express that they have a **complaint** to make.

a) If the chatbot fails to understand three consecutive times:

→ The chatbot automatically **suggests contacting** a theater **employee**.

- It provides a button that **redirects** the user to the **theater info screen**, where further **options** are **available**.
- The user **presses** the button.

3a) Contact Screen

On the **theater info screen**, the user clicks the **phone icon** to make a **direct call** to a **theater employee** for **assistance**.

b) If the chatbot understands the complaint:

The **chatbot** explicitly provides a **button** to **redirect** the user to the "**Complaint**" **screen** or make the complaint **through** the **chat**.

I. The user presses this button.

3b) Make Complaint screen:

The **user** **fills** out the **requested information** in the **text fields** and specifies their **complaint** using options from the **drop-down menu**.

→ After **submitting** the **complaint**, the **app** **confirms receipt** with a **message** such as:

“Thank you for submitting your complaint. Our team will review it and get back to you as soon as possible.”

II. The user chooses to stay in the chat screen and make the complaint through it.

The **user** through **screech** or **text** explains through **messages** to the **assistant** the **problem/complaint**.

→ The **chatbot** interacts **naturally** with the **user** to **submit** the **complaint**.

→ After **submitting** the complaint, the **chatbot** **confirms receipt** with a **message** such as:

“Thank you for submitting your complaint. Our team will review it and get back to you as soon as possible.”

The walkthrough **demonstrates** that **users**, even those **unfamiliar** with the application, can **smoothly** **navigate** the **complaint submission** process with **confidence**. The interface **guides** users **step by step**, **minimizing ambiguity** and **supporting clear, intentional actions**. From launching the **chatbot** to final **confirmation** a **logical progression** underlies, which makes the experience **accessible** and **stress-free**.

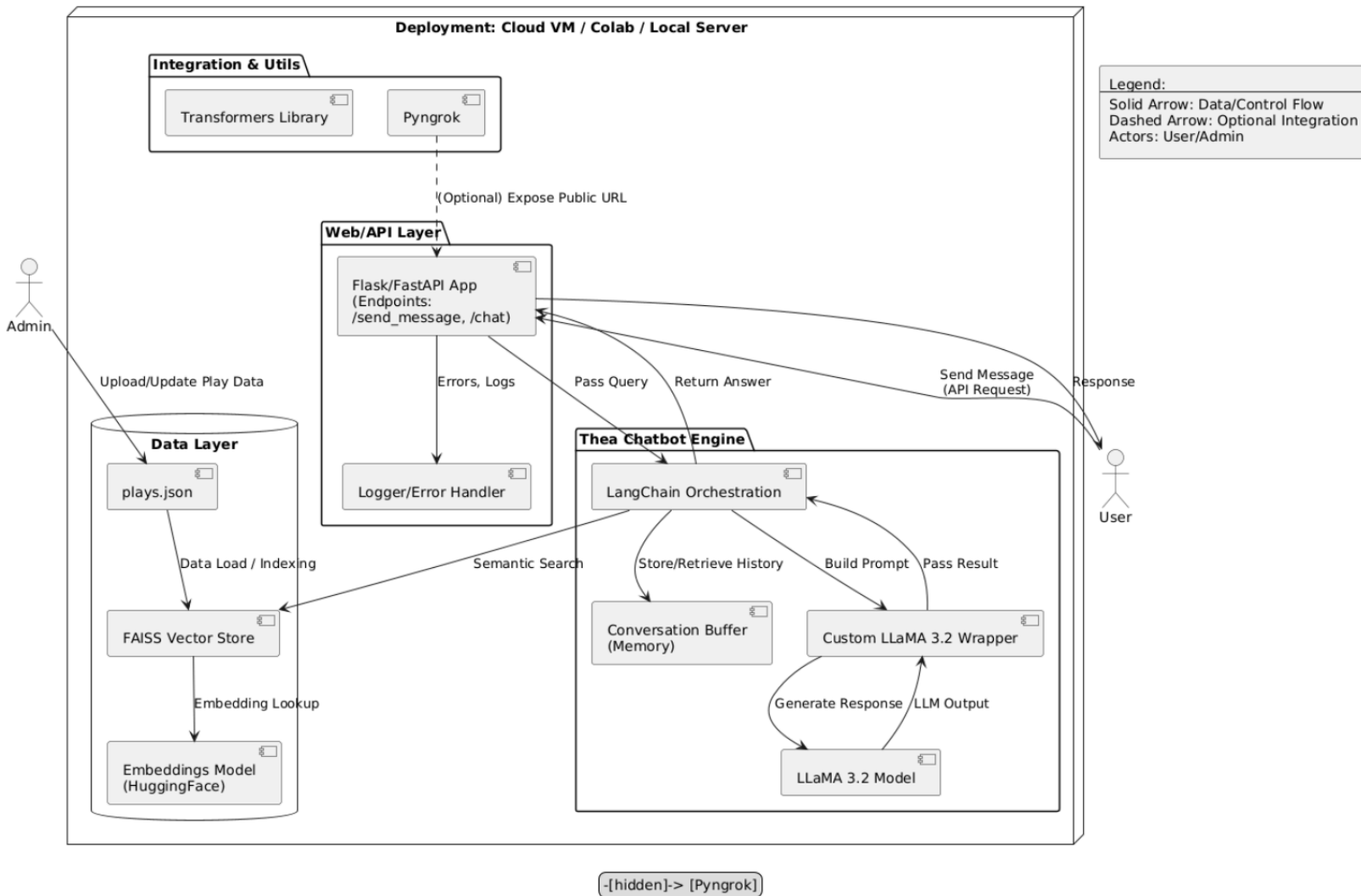
The system **significantly reduces** the “*gulf of execution*”, as users are **never left guessing** what to do **next**. **Buttons** are **well-labeled**, instructions are **concise**, and every available **action** matches standard user **mental models**. Entering **text** or selecting **options from drop-down menus** feels **familiar** and **straightforward**, further **lowering barriers** for all **user groups**.

Likewise, the “*gulf of evaluation*” is kept to a **minimum**. The application **immediately responds** to **user inputs** with **clear feedback**, so users **always know** the **outcome** of their **actions**. If the **chatbot** encounters **difficulty understanding the complaint**, it **transparently offers alternatives**, such as **connecting to a human representative** or **redirecting to a dedicated complaint form**. When the process is **successful**, a **visible confirmation** reassures the user that their request is being **handled**.

In summary, the interface is **highly supportive** of **users’ goals** and **encourages completion** of **tasks** with **little to no confusion**. The **combination** of **guided interactions**, **responsive feedback**, and **accessible language** ensures that even **users** with **limited technical skills** can **complete the complaint process efficiently**. This highlights the **strong usability** and **human-centered design** at the **core of the application**.

LLM Chatbot with RAG Pipeline Explanation

Architecture Diagram: Thea Conversational AI Platform



(Flask + LLaMA 3.2 + LangChain + FAISS)